

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

Two-variable data: Models and scatterplots

02

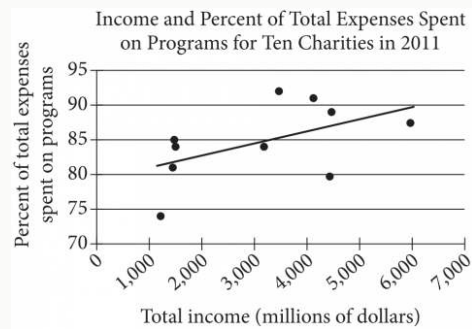
10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

An inspector begins a day of work with a large sample of shirts that need to be checked for defects. The inspector works at a constant rate throughout the morning. What type of model is best to model the number of shirts remaining to be checked for defects at any given time throughout the morning?

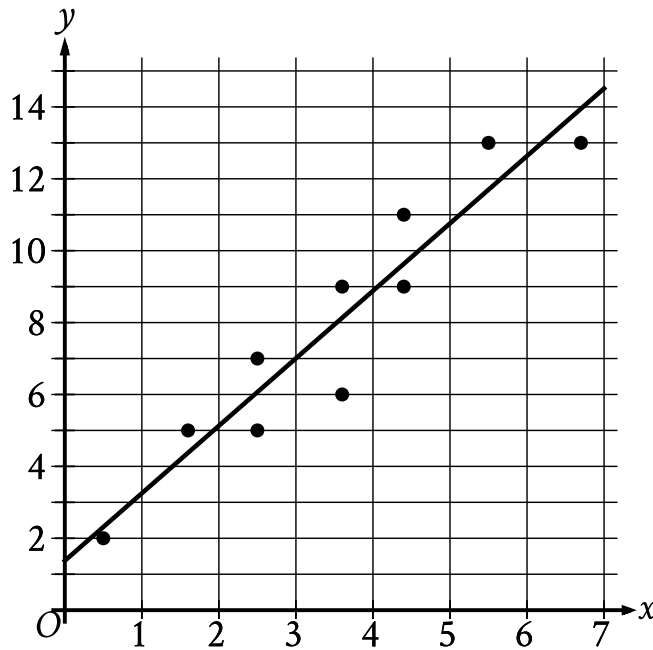
- A. A linear model with a positive slope
- B. A linear model with a negative slope
- C. An exponential growth model
- D. An exponential decay model



The scatterplot above shows data for ten charities along with the line of best fit. For the charity with the greatest percent of total expenses spent on programs, which of the following is closest to the difference of the actual percent and the percent predicted by the line of best fit?

- A. 10%
- B. 7%
- C. 4%
- D. 1%

In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants up from left to right.
 - The line of best fit passes through the following approximate coordinates:
 - (1 comma 3.3)
 - (3 comma 7.0)
 - (5 comma 10.8)

Which of the following is closest to the slope of the line of best fit shown?

A. 0

B. $\frac{1}{2}$

C. 1

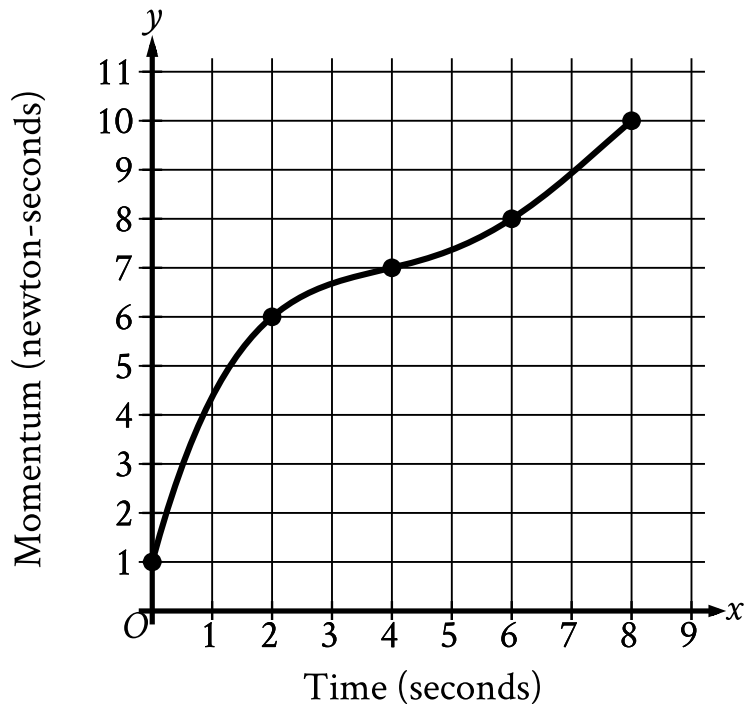
D. 2

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

Which of the following is true about the values of 2^x and $2x+2$ for $x > 0$?

- A. For all $x > 0$, it is true that $2^x < 2x+2$.
- B. For all $x > 0$, it is true that $2^x > 2x+2$.
- C. There is a constant c such that if $0 < x < c$, then $2^x < 2x+2$, but if $x > c$, then $2^x > 2x+2$.
- D. There is a constant c such that if $0 < x < c$, then $2^x > 2x+2$, but if $x > c$, then $2^x < 2x+2$.



- In quadrant 1:
 - The curve begins at point (0 comma 1).
 - The curve rises sharply to point (2 comma 6).
 - The curve rises gradually to point (4 comma 7).
 - The curve rises gradually to point (6 comma 8).
 - The curve rises sharply and ends at point (8 comma 10).

The graph shows the momentum y , in newton-seconds, of an object x seconds after the object started moving, for $0 \leq x \leq 8$. What is the average rate of change, in newton-seconds per second, in the momentum of the object from $x = 2$ to $x = 6$?

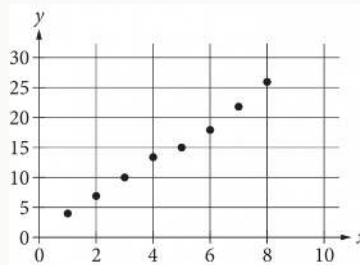
STUDENT-PRODUCED RESPONSE

2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

PROBLEM-SOLVING AND DATA ANALYSIS



Which of the following could be the equation for a line of best fit for the data shown in the scatterplot above?

- A. $y = 3x + 0.8$
- B. $y = 0.8x + 3$
- C. $y = -0.8x + 3$
- D. $y = -3x + 0.8$

In which of the following tables is the relationship between the values of x and their corresponding y -values nonlinear?

A.

x	1	2	3	4
y	8	11	14	17

B.

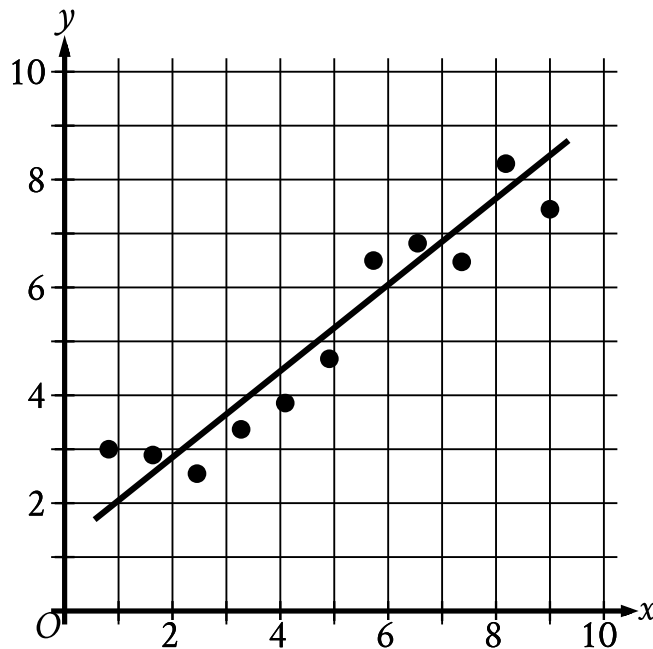
x	1	2	3	4
y	4	8	12	16

C.

x	1	2	3	4
y	8	13	18	23

D.

x	1	2	3	4
y	6	12	24	48



- The scatterplot has 11 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants up from left to right.
 - 5 points are above the line of best fit.
 - 6 points are below the line of best fit.
 - The line of best fit passes through the following approximate coordinates:
 - (2 comma 2.9)
 - (8 comma 7.7)

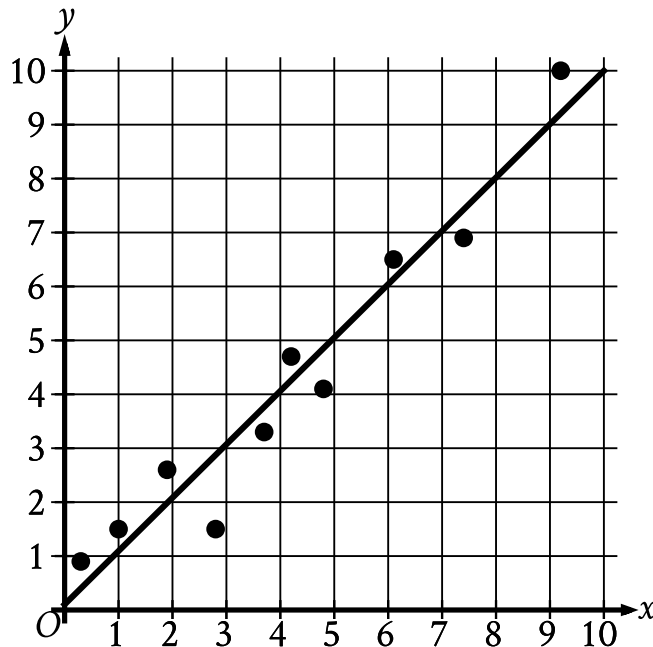
The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown. For how many of the 11 data points does the line of best fit predict a greater y -value than the actual y -value?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The scatterplot shows the relationship between two variables, x and y . A line of best fit for the data is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants up from left to right.
 - 6 points are above the line of best fit.
 - 4 points are below the line of best fit.
 - The line of best fit passes through the following approximate coordinates:
 - (0 comma 0.1)
 - (5 comma 5.1)

For how many of the 10 data points is the actual y -value greater than the y -value predicted by the line of best fit?

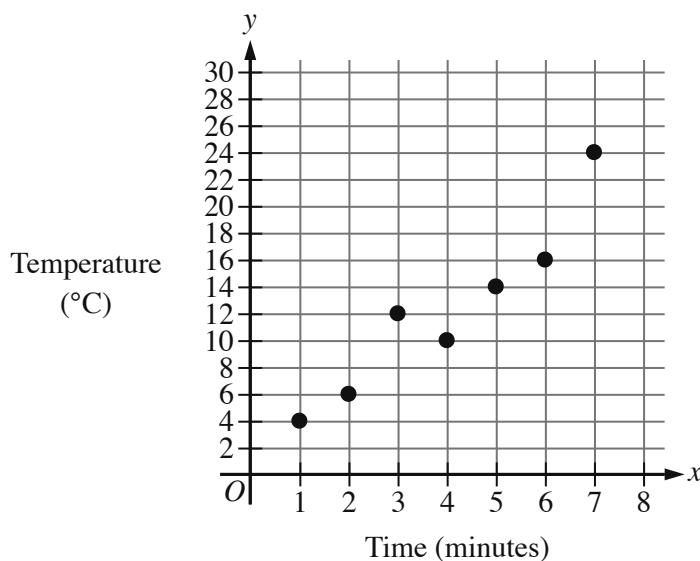
A. 3

B. 4

C. 6

D. 7

During a study, the temperature, in degrees Celsius $^{\circ}\text{C}$, of the air in a chamber was recorded to the nearest integer at certain times. The scatterplot shows the recorded temperature y , in $^{\circ}\text{C}$, of the air in the chamber x minutes after the start of the study.



- The scatterplot has 7 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
 - (1 comma 4)
 - (2 comma 6)
 - (3 comma 12)
 - (4 comma 10)
 - (5 comma 14)
 - (6 comma 16)
 - (7 comma 24)

What was the average rate of change, in $^{\circ}\text{C}$ per minute, of the recorded temperature of the air in the chamber from $x = 5$ to $x = 7$?

STUDENT-PRODUCED RESPONSE

☐ .	☐ /	☐ .	☐ /	☐ .
☐ 0	☐ 0	☐ 0	☐ 0	☐ 0
☐ 1	☐ 1	☐ 1	☐ 1	☐ 1
☐ 2	☐ 2	☐ 2	☐ 2	☐ 2
☐ 3	☐ 3	☐ 3	☐ 3	☐ 3
☐ 4	☐ 4	☐ 4	☐ 4	☐ 4
☐ 5	☐ 5	☐ 5	☐ 5	☐ 5
☐ 6	☐ 6	☐ 6	☐ 6	☐ 6
☐ 7	☐ 7	☐ 7	☐ 7	☐ 7
☐ 8	☐ 8	☐ 8	☐ 8	☐ 8
☐ 9	☐ 9	☐ 9	☐ 9	☐ 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.